

EDMONTON NAMAQ, AB, Canada

WMO: 711210

Lat: 53.6639N

Lon: 113.4661W

Elev: 2257

StdP: 13.54

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-16.1	-30.9	1.0	-21.3	-25.0	1.5	-15.8	24.9	21.8	22.1	21.5	5.7	270	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	82.0	63.9	79.0	62.4	76.0	60.8	66.3	78.3	64.4	76.0	62.5	73.2	9.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	89.9	72.3	59.7	83.3	69.6	57.9	78.0	67.6	32.3	78.4	30.7	76.0	29.3	73.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.6	19.5	16.9		-29.6	87.6	6.5	3.2	-34.2	89.9	-38.1	91.8	-41.7	93.6	-46.4	96.0	
				-29.8	70.4	6.3	2.3	-34.3	72.1	-37.9	73.4	-41.5	74.7	-46.0	76.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.7	13.2	13.5	24.2	38.6	50.9	58.2	62.6	60.2	52.1	39.5	24.7	13.5
	DBStd	21.26	15.50	14.45	13.82	10.19	8.28	5.63	5.39	5.71	7.57	8.73	12.02	14.09
	HDD50	5713	1140	1021	801	358	89	6	1	3	63	339	760	1132
	HDD65	10027	1605	1441	1266	793	440	216	111	166	390	792	1210	1597
	CDD50	1231	0	0	0	16	117	250	390	320	126	12	0	0
	CDD65	70	0	0	0	0	3	11	36	18	2	0	0	0
	CDH74	1074	0	0	0	3	87	164	439	283	95	3	0	0
	CDH80	189	0	0	0	0	11	25	94	43	16	0	0	0
Wind	WSAvg	7.9	7.7	7.6	8.1	9.2	9.3	8.1	7.3	6.8	7.5	8.1	7.6	7.4
Precipitation	PrecAvg	16.5	0.7	0.5	0.6	1.1	1.9	2.7	3.5	2.1	1.1	0.8	0.6	0.7
	PrecMax	21.8	3.3	1.2	1.9	1.9	3.9	5.7	5.7	4.5	3.0	2.1	1.3	1.3
	PrecMin	9.3	0.1	0.2	0.2	0.1	0.3	0.8	0.7	0.7	0.0	0.2	0.1	0.2
	PrecStd	3.1	0.7	0.2	0.4	0.5	0.9	1.2	1.2	1.0	0.7	0.5	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	45.5	54.7	72.8	81.4	82.9	87.7	84.3	82.5	71.1	54.4	44.8
		MCWB	36.4	37.5	42.6	52.2	58.0	63.0	67.9	64.6	61.3	54.4	43.9	37.2
	2%	DB	40.9	40.3	48.6	66.2	76.1	78.6	82.3	80.6	76.3	62.2	47.2	39.8
		MCWB	34.6	33.5	38.3	48.4	55.4	60.5	65.9	64.0	59.0	48.9	39.4	33.6
	5%	DB	37.7	36.8	44.2	60.4	71.9	74.7	79.0	77.2	71.7	57.2	43.0	36.4
		MCWB	32.5	31.4	36.3	45.3	53.3	58.8	64.2	63.1	57.2	46.0	36.7	31.2
	10%	DB	34.1	33.5	40.6	55.3	67.4	71.2	75.4	73.8	66.5	52.9	39.3	32.7
		MCWB	29.9	29.2	34.1	42.7	51.0	57.0	62.4	61.4	54.4	43.4	34.4	28.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	37.7	43.0	52.6	59.5	65.5	70.2	68.0	62.9	54.9	44.1	37.6
		MCDB	43.1	45.1	53.5	70.7	77.6	78.9	81.6	78.5	78.4	69.0	52.4	44.4
	2%	WB	35.0	34.0	39.0	49.3	56.6	62.3	67.4	65.6	60.2	50.0	40.2	34.0
		MCDB	40.3	39.4	47.1	64.8	72.2	74.9	79.4	77.3	74.5	60.6	46.4	38.8
	5%	WB	32.7	31.8	36.5	46.1	54.3	60.2	65.3	63.8	57.7	46.8	37.3	31.5
		MCDB	37.5	36.5	43.6	59.3	69.8	71.9	76.7	75.2	70.0	55.3	42.0	36.2
	10%	WB	29.8	29.1	34.2	43.3	52.0	58.4	63.3	62.0	55.3	44.4	34.8	28.5
		MCDB	34.1	33.6	40.1	54.1	65.3	68.8	73.6	72.3	65.0	51.6	38.8	32.7
Mean Daily Temperature Range	5% DB	MDBR	15.9	17.3	17.3	19.9	22.5	20.3	20.4	21.8	21.9	19.0	15.2	15.5
		MCDBR	17.1	20.0	20.7	29.3	29.3	26.4	26.3	27.9	30.5	26.5	19.4	17.2
		MCWBR	13.1	15.2	13.8	16.3	14.5	12.6	13.1	13.9	15.9	15.6	13.2	13.4
	5% WB	MCDBR	16.9	19.2	19.9	28.1	26.9	23.1	24.3	25.6	28.9	25.0	18.1	17.3
	MCWBR	13.5	14.9	13.6	16.3	13.6	13.6	11.9	13.0	13.5	15.3	15.2	12.9	13.8
Clear-Sky Solar Irradiance	taub	0.256	0.266	0.311	0.325	0.341	0.352	0.354	0.361	0.307	0.281	0.253	0.239	
	taud	2.307	2.354	2.303	2.391	2.395	2.381	2.419	2.352	2.502	2.549	2.456	2.315	
	Ebn at Noon	232	270	277	288	287	285	282	272	278	264	238	217	
	Edh at Noon	18	24	31	33	35	36	34	34	26	20	16	15	
All-Sky Solar Radiation	RadAvg	305	634	1124	1482	1816	1909	1950	1604	1107	635	303	224	
	RadStd	34	49	86	139	135	124	95	111	114	76	34	21	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page